Walchand Institute of Technology, Solapur
(An Autonomous Institute)

Affiliated to
Punyashlok Ahilyadevi Holkar Solapur University,
Solapur

Choice Based Credit System (CBCS)

Structure and Syllabus
for
M. Tech. in Computer Science and Engineering
As per NEP 2020

F.Y.M. Tech. Computer Science and Engineering W.E.F. 2026-27

Dean (Academics)

HEAD
Computer Sc. and Engg. Deptt
W. I. T., Solapur

WIT SOLAPUR (Autonomous) WALCHAND INSTITUTE OF TECHNOLOGY

Computer Science and Engineering

Vision

To develop professional engineers in Computer Science & Engineering having ethical values, research aptitude and ability to address challenges of modernization in the IT industry aiming at overall sustainable development of the society.

Mission

- M1 - To impart quality education in the field of Computer Science & Engineering in accordance with the needs of the Modernization & Globalization through technology enabled education.
- M2 - To inculcate lifelong learning in students to face challenges posed by ever-changing IT career landscape as a disciplined professional with a sense of professional ethics.
- M3 - To inculcate critical thinking and creativity for identifying various societal issues and to provide solutions.
- M4 - To enhance career opportunities for students through academia-industry interaction and research.

Computer Science and Engineering

Post Graduate Programme

Program Educational Objectives (PEOs)

1. Graduates will exhibit the technical knowledge to provide innovative solutions for real-time problems.
2. Graduates will develop research aptitude and the lifelong learning to address the challenges in IT industries.
3. Graduates will pursue leadership skills, professionalism and ethical values during their career for the growth of society.

Program Outcomes (POs)

Postgraduates on successful completion of the programme will be able to:

PO1	Independently carryout research/investigation and development work to solve practical problems.
PO2	Write and present a substantial technical report/document.
PO3	Demonstrate a degree of mastery in Computer Science and Engineering; a level higher than the requirements in the appropriate bachelor program.
PO4	Analyze, evaluate and build systems in the field of Computer Science & Engineering.
PO5	Practice ethical behavior with professional code of conduct, effective managerial skills and life- long learning for solving real world problems related to social and environmental up liftment.
PO6	Acquire expertise in the latest trends in Computer Science & Engineering like data science, computer security and advanced computing leading to innovation.
PO7	Ability of life-long self learning and to keep one self up-to-date in the field of technology.

Legends used–

L	Lecture
T	Tutorial
P	Practical
D	Drawing
FA	Formative Assessment
SA	Summative Assessment
ESE	End Semester Examination
ISE	In Semester Evaluation
ICA	Internal Continuous Assessment
F.Y.	First Year
S.Y.	Second Year
M.Tech.	Master of Technology

Course Codes	
CC	Core Compulsory Course
SN*	Self-Learning **N***indicates the serial number of electives offered in the respective category*
EN*	Core Elective **N***indicates the serial number of electives offered in the respective category*
SM	Seminar
MP	Mini Project
PR	Project

WALCHAND INSTITUTE OF TECHNOLOGY

STRUCTURE OF M.Tech.(COMPUTER SCIENCE AND ENGINEERING)

Choice Based Credit System Syllabus w.e.f. 2026-27

Structure–(Scheme 24)

Semester-I

Course Code	Name of the Course	Engagement Hours			Credits	FA	SA		Total
		L	T	P		ESE	ISE	ICA	
CSP1CC1	Applied Algorithms	3	--	2	4	60	40	25	125
CSP1CC2	Theory of Computation	3	--	--	3	60	40	--	100
CSP1CC3	Internet of Things	3	--	--	3	60	40	--	100
CMP1RM	Research Methodology and IPR	3	1	--	4	60	40	25	125
CSP1CE1N*	Core Elective I	3	--	2	4	60	40	25	125
CSP1SM1	Seminar-I	--	--	4	2	--	--	50	50
	Total	**15**	**1**	**8**	**20**	**300**	**200**	**125**	**625**

N*indicates course serial number of elective offered in the respective category

Course Code	Name of the Course	Engagement Hours			Credits	FA	SA		Total
		L	T	P		ESE	ISE	ICA	
CMP1AC	Yoga for Stress Management	2	--	--	--	–	50	--	50

Core Elective-I

Course Code	Course Title
CSP1CE11	Data Mining
CSP1CE12	Advanced Cryptography
CSP1CE13	Natural Language Processing

Note:L-Lectures,P-Practical,T-Tutorial,ISE-InSemesterEvaluation,ESE-EndSemesterEvaluation, ICA-Internal Continuous Assessment

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STRUCTURE OF M.Tech.(COMPUTER SCIENCE AND ENGINEERING)

Choice Based Credit System Syllabus w.e.f. 2026-27

Semester-II

Course Code	Name of the Course	*Engagement Hours*			*Credits*	FA	SA		*Total*
		L	*T*	*P*		ESE	*ISE*	*ICA*	
CSP2CC4	Advanced Cloud Computing	3	--	--	3	60	40		100
CSP2CC5	Internet Routing Algorithm	3	--	--	3	60	40		100
CSP2CC6	Advanced Database Systems	3	--	2	4	60	40	25	125
CSP2CE2N*	Core Elective-II	3	--	2	4	60	40	25	125
CSP2CE3N*	Core Elective–III	3	--	--	3	60	40	--	100
CSP2SM2	Seminar-II	--	--	6	3	--	--	100	100
	Total	**15**	--	**12**	**20**	**300**	**200**	**150**	**650**

Note:-Students have to compulsorily undergo internship of one month after the Semester-II(in the summer vacation). The evaluation of the same will be carried out at the end of semester – III.

Core Elective-II

Course Code	Course Title
CSP2CE21	Machine Learning
CSP2CE22	BlockChain Technology
CSP2CE23	Reinforcement Learning

Core Elective-III

Course Code	Course Title
CSP2CE31	Data Science
CSP2CE32	Wireless Sensor Network
CSP2CE33	High Performance Computing

Note:T-Lectures,P-Practical,A-Tutorial,ISE-InSemesterEvaluation,ESE-EndSemesterEvaluation, ICA-Internal Continuous Assessment

- Student shall deliver all seminars using modern presentation tools. A hard copy of the report shall be submitted to the department before delivering the seminar. A PDF copy of the report must be submitted to the advisor along with other details if any.

Dean (Academics)

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WALCHAND INSTITUTE OF TECHNOLOGY

WALCHAND INSITUTE OF TECHNOLOGY

M.Tech.(Computer Science and Engineering)Part I, Semester I

CSP1CC1-APPLIED ALGORITHMS

Teaching Scheme	**Examination Scheme**
Lectures- 3 Hours/week, 3Credits	**ESE–60 Marks**
Practical- 2 Hour/week, 1 Credit	**ISE–40 Marks**
	ICA–25 Marks

COURSE OUTCOMES:

At the end of the course, students will be able to

1. Select appropriate algorithm design paradigm for a problem.
2. Apply Graph algorithm for a real time problem.
3. Design and implement Approximation and Probabilistic algorithms.

SECTION-I

Unit 1 - Foundations **(6)**

Algorithms, Performance of algorithms, Growth of Functions-Asymptotic notation, Amortized analysis, Solving recurrences- Substitution method, Master method

Unit 2 – Graph Algorithms **(7)**

Minimum spanning tree -Prim"s and Krushkal"s Algorithm for, Single-Source Shortest Paths-The Bellman-Ford algorithm and Dijkstra"s algorithm, Maximum Flow-Flow networks, The Ford-Fulkerson Algorithm, Huffman codes.

Unit 3 – Dynamic Programming **(7)**

Matrix-chain multiplication, longest common subsequences, All-Pairs Shortest Paths, The Floyd-Warshall algorithm, Johnson Algorithm, optimal binary search tree, Reliability Design.

SECTION-II

Unit 4 - Computational Geometry **(8)**

Prerequisites-Basic properties of line, intersection of line, linesegment, polygon etc. Line segment properties, detaining segment intersection in time complexity, Convex full problem – formulation,solvingbyGrahamscanalgorithm,Jarvismarchalgorithm,closestpairofpoints

Unit 5 NP-Completeness and Approximation Algorithms **(7)**

NP-Completeness:NP-completeness and reducibility, NP-completeness proofs,NP-complete problems, Approximation algorithms: The vertex-cover problem, The traveling-salesman problem, The set covering problem, The subset-sum problem

Unit 6 – Applied Algorithms **(8)**

Number-Theoretic: Number Theoretic notion, Greatest common divisor, The Chinese remainder theorem, RSA. String Matching Algorithms: The Rabin-Karp algorithm, The Knuth-Morris-Pratt algorithm. Parallel Algorithm: Mesh Algorithm and its applications. Probabilistic Algorithm: Game Theoretic Techniques. Randomized Algorithms: Definition, Monte Corlo and Las Vegas algorithms

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Internal Continuous Assessment (ICA):

Minimum 5 to 6 assignments based on above topics.

Text Books:

1. Ellis Horowitz, Sartaj Sahni – Fundamental of Computer Algoritms,Universities Press,II Edition
2. Bressard, Bratley-Fundamental of Algorithms, PHI, 2nd Edition
3. Thomas H. Cormen and Chales E.L. Leiserson, Introduction to Algorithm ,PHI, 2nd Edition

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Reference Books

1.A.V.Aho and J.D.Ullman, Design and Analysis of Algorithms, Addition Wesley, 2nd Edition

WALCHANDINSTITUTEOFTECHNOLOGY
M.Tech. (Computer Science & Engineering) Part I, Semester I
CSP1CC2 - THEORY OF COMPUTATION

Teaching Scheme	**Examination Scheme**
Lectures - 3 Hours/week, 3Credits	**ESE - 60 Marks**
	ISE - 40 Marks

COURSE OUTCOMES:

At the end of the course students will be able to,

1. Design deterministic and non-deterministic finite automata, context-free grammars and pushdown automata.
2. Design and analyze Turing machines, their capabilities and limitations.
3. Demonstrate the understanding of complexity classes and unsolved problems in theoretical computer science.
4. Apply the theoretical concepts to the practice of program design with regular expressions, parsing, decidability and complexity analysis.

SECTION-I

Unit 1 - Introduction: (8)

Introduction To Finite Automata: Alphabets and languages- Deterministic Finite Automata – Non-deterministic Finite Automata–Equivalence of Deterministic and Non-Finite Automata Languages Accepted by Finite Automata – Finite Automata and Regular Expressions – Properties of Regular sets & Regular Languages and their applications. Context Free Languages: Context – Free Grammar– Regular Languages and Context-Free Grammar – Pushdown Automata – Pushdown Automata and Context-Free Grammar – Properties of Context-Free Languages– pushdown automata and Equivalence with Context Free Grammars

Unit 2 - Turing machine: (5)

Turing machines,variants of TMs, programming techniques for TMs,TMs and computers.

Unit 3 – Decidability : (5)

Decidable languages, decidable problems concerning Context-free languages. The halting problem – Diagonalization method, halting problem is undecidable.

SECTION II

Unit 4 - Reducibility: (6)

Undecidable problems from language theory, Regular expressions, Turing machines, Reduction, A simple undecidable problem(PCP),mapping reducibility and other undecidable problems.

Unit 5 - Computability: (6)

Primitive recursive functions, more examples, The recursion theorem.

Unit 6 – Computational complexity: (6)

Tractable and Interactable problems, Growth rates of functions, Time complexity of TM, Tractable decision problems, Theory of Optimization.

Text Books:

1. Introduction to Theory of Computation-Michael Sipser (Thomson Brools Cole)
2. Introduction to Automata Theory, Languages and Computation- J.E.Hoperoft, Rajeev Motawani and J.D.Ullman (Pearson Education Asia) 2nd Edition.
3. Theory of Computer Science–E.V. Krishnamoorthy
4. Introduction to languages & theory of computation – John C.Martin (MGH)

References:

1. Theory of Computation-A Problem Solving Approach - Kavi Mahesh (Wiley India)
2. Theory of Computation -Dr.O.G. Kakde (University Science Press)
3. Formal Languages &Automata Theory- Basavraj S.Anami, Karibasappa K.G.,Wiley Precise Textbook-Wiley India
4. Theory of Computation -Rajesh K Shukla (CENGAGE Learning)

WALCHAND INSTITUTE OF TECHNOLOGY
M.Tech.(Computer Science & Engineering) Part I, Semester I
CSP1CC3-RESEARCH METHODOLOGY and IPR

Teaching Scheme	**Examination Scheme**
Lectures -3 Hours/week,3 Credits	**ESE - 60 Marks**
Tutorials - 1 Hour/week, 1 Credit	**ISE - 40 Marks**
	ICA - 25 Marks

Course Outcomes: -
At the end of the course, student will be able to,
1. Propose and distinguish appropriate research designs and methodologies for a specific research project.
2. Develop skills in literature review, qualitative and quantitative data analysis and presentation.
3. Describe the importance of Computers, Information Technology in research and also highlight the significance of ideas, concept, and creativity in research.
4. Illustrate the importance of Intellectual Property Rights in growth of individuals & nation.
5. Exhibit knowledge about IPR protection, providing an incentive to inventors for further research work leading to creation of new and better products.

SECTION-I

Unit 1 - Introduction: (6)
Defining Research, Scientific Enquiry, Hypothesis, Scientific Method, Types of Research, Research Process and steps in it. Research Proposals – Types, contents, sponsor agent's requirements, Ethical, Training, Cooperation and Legal aspects.

Unit 2 – Research Design: (6)
Meaning, Need, Concepts related to it, categories; Literature Survey and Review, Dimensions and issues of Research Design, Research Design Process – Selection of type of research, Measurement and measurement techniques ,Selection ofSample, Selection of Data Collection Procedures, Selection of Methods of Analysis, Errors in Research.

Unit 3 – Research Problem: (6)
Problem Solving – Types, Process and Approaches – Logical, Soft System and Creative; Creative problem solving process, Development of Creativity, Group Problem Solving Techniques for Idea Generation – Brain storming and Delphi Method.

SECTION-II

Unit 4 – Nature of Intellectual Property: (8)
Patents, Designs, Trade marks and Copyright. Process of Patenting and Development: Technological research, Innovation, Patenting, Development. International Scenario: International cooperation on Intellectual Property. Procedure for grants of patents, Patenting under PCT.

Unit 5 - Patent Rights: (5)
Scope of Patent Rights. Licensing and transfer of technology. Patent information and databases. Geographical Indications.

Unit 6 – New Developments in IPR: (5)
Administration of Patent System. New developments in IPR; IPR of Biological Systems, Computer Software etc. Traditional knowledge Case Studies, IPR

Internal Continuous Assessment(ICA):
Minimum 6 to 8 assignments on above topics.

References:

1. Vinayak Bairagi, Mousami V. Munot, Vinayak Bairagi, Mousami V. Munot, Research Methodology, A Practical and Scientific Approach, Publisher: Chapman and Hall/CRC ,Year 2019, ISBN 9781351013253.Krishnaswamy, K.N., Sivakumar, AppaIyer & Mathirajan M., (2006) - Management Research Methodology: Integration of Principles, Methods & Techniques (New Delhi, Pearson Education)
2. Montgomery, Douglas C. (2004) – Design & Analysis of Experiments, (New York, John Wiley &Sons)
3. Kothari, C.K. (2004) – Research Methodology, Methods & Techniques, (New Delhi, New Age International Ltd. Publishers).
4. Prabuddha Ganguli, IPR: Unleashing the Knowledge Economy, published by Tata McGraw Hill2001.
5. John W Cresswell, (2009)-Research Design: Qualitative, Quantitative and Mixed Methods Approaches, (Sage Publications Pvt Ltd. 3rd Edition.)
6. RanjitKumar,2nd Edition, “Research Methodology: A Step by Step Guide for beginners”
7. Halbert,“Resisting Intellectual Property”,Taylor & Francis Ltd,2007.
8. Robert P. Merges, Peter S. Menell, Mark A. Lemley, “Intellectual Property in NewTechnological Age”, 2016.

WALCHAND INSTITUTE OF TECHNOLOGY
M. Tech. (Computer Science & Engineering) Part I, Semester I
CSP1CC4 – INTERNET OF THINGS

Teaching Scheme
Lectures–3 Hours/week, 3 Credits

Examination Scheme
ESE – 60 Marks
ISE – 40 Marks

COURSE OUTCOMES:
At the end of the course students will be able to
1. Interpret the characteristics and applications of IoT for deployment of the architectural model.
2. Compare smart objects and associated technologies for deployment in the network.
3. Analyze and choose the IoT protocol for efficient network communication.
4. Apply security concerns and challenges while implementing IoT solutions.
5. Provide the appropriate IoT solutions to the given problem.

SECTIONI

Unit 1 - OVERVIEW OF IOT **(6)**
Introduction – Design Principles for connected Devices –Prototyping for embedded devices-Prototyping for Physical design, Characteristics of IoT, Sensornetworks, Applications of IoT

Unit - 2 IOT ARCHITECTURE **(6)**
Node Structure-Sensing-Processing - Communication - Powering - Networking - Topologies-Layer/Stack architecture - IoT Standards- Cloud computing for IoT-Bluetooth-Bluetooth Low Energy- beacons.

Unit 3 – WIRELESS TECHNOLOGY FOR IOT **(7)**
WiFi (IEEE 802.11) - Bluetooth/Bluetooth Smart - ZigBee/ZigBee Smart - UWB (IEEE802.15.4)-6LoWPAN - Proprietary systems.

SECTION II

Unit 4 – BUILDING IOT WITH RASPBERRY PI **(6)**
RASPBERRY PI:Physical device-Raspberry Pi Interfaces–Programming-APIs/Packages- Web services

Unit 5 – Database implementation for IoT **(7)**
Cloud based IoT platforms, SQL vs. NoSQL, Open sourced vs. Licensed Database, Available M2M cloud platform, AxedaXively, Omega NovoTech, AylaLibellium, CISCOM2M platform, AT &T M2M platform, Google M2M platform

Unit 6 - CASESTUDIES **(6)**
Home Automation-smart cities-Smart Grid- Electric vehicle charging- Environment-Agriculture-Productivity Applications

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Text Books:

1. Adrian McEwen and Hakim Cassimally "Designing the Internet of Things", Wiley,2014.(UNIT I & V)
2. Oliver Hersent , David Boswarthick and Omar Elloumi "The Internet of Things",Wiley,2016 (UNIT II, III & UNIT V)
3. PeterWaher,"Learning Internet of Things", Packt Publishing,2015(UNIT IV)
4. Iot-Enabled Applications Third Edition,by Gerardus Blokdyk(Author)

Reference Books:

1. Jean-Philippe Vasseur, Adam Dunkels, "Interconnecting Smart Objects with IP:The Next Internet" Morgan Kuffmann Publishers, 2010
2. Arshdeep Bahga and Vijai Madisetti : A Hands-on Approach "Internet of Things", Universities Press 2015.
3. Samuel Greengard,"The Internet of Things", The MIT press, 2015
4. Ovidiu Vermesan and Peter Friess(Editors), "Internet of Things : Converging Technologies for Smart Environments and Integrated Ecosystems", River Publishers Series in Communication,2013
5. https://www.coursera.org/specializations/iot

WALCHAND INSTITUTE OF TECHNOLOGY

M.Tech.(Computer Science and Engineering) Part I, Semester I

CSP1CE11 : CORE ELECTIVE-I : DATAMINING

Teaching Scheme	**Examination Scheme**
Lectures–3 Hours/week, 3 Credits	**ESE–60Marks**
Practical– 2 Hours/week, 1 Credit	**ISE –40 Marks**
	ICA–25Marks

COURSE OUTCOMES:

After under going the course, Students will be able to:

1. Comprehend data mining principles and techniques.
2. Design a datamart or data warehouse using multidimensional schemas.
3. Identify business applications of data mining and extract knowledge using data mining techniques.
4. Develop skills to write queries using DMQL
5. Adapt to new data mining tools and trends.

SECTION-I

Unit - 1: Introduction **(3)**

DataWarehousingandIntroductiontodataminingbasicelementsofdatawarehousing,Data warehousing and OLAP.

Unit - 2 : Data model development for Data Warehousing: **(3)**

Businessmodel,selectionofthedataofinterest,creationandmaintainingkeys,modelingtransaction,data warehousing optimization.

Unit - 3:Data warehousing methodologies **(3)**

Type and comparisons.

Unit – 4 : Data Mining techniques **(6)**

Data mining algorithms, classification, Decision-Treebased Classifiers clustering, association Association-Rule Mining Information Extraction using Neural Networks.

Unit – 5 : Knowledge discovery **(3)**

KDD environment

SECTION-II

Unit – 6 :Visualization **(4)**

Data generalization and summarization-based characterization, Analytical characterization: analysis of attribute relevance, mining class Comparison, Discriminating between classes, mining descriptive statistical measures in large database.

Unit - 7 : Data mining primitives, languages & system architectures **(4)**

Data mining primitives, Query language, designing GUI based on a data mining query language, architectures of data mining systems.

Unit – 8 : Advanced topics **(3)**

Spatial mining, temporal mining.

Unit – 9 : Web mining **(3)**

Web content mining,web structure mining, web usage mining

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Unit – 10 : Application and trends in data mining **(4)**

Applications, systems products and research prototypes, multimedia data mining, indexing of multimedia material, compression, space modeling.

Internal Continuous Assessment (ICA):

Minimum 5 to 6 assignments based on above topics.

Textbooks:

1. Paulraj Ponniah – Web warehousing fundamentals - JohnWiley.
2. M.H.Dunham - Data mining introductory and advanced topics – Pearson education
3. Han, Kamber - Data mining concepts and techniques, Morgan Kaufmann

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Reference Book:

Imhoff, Galemmo, Geiger – Mastering data warehouse design, Wiley DreamTech

CSP1CE12: CORE ELECTIVE-I :ADVANCED CRYPTOGRAPHY

Teaching Scheme	**Examination Scheme**
Lectures - 3 Hours/week, 3 Credits	**ESE–60Marks**
Practical - 2 Hours/week, 1 Credit	**ISE –40 Marks**
	ICA–25Marks

COURSE OUTCOME:

1. Describe how various cryptography algorithms and protocols work.
2. Apply crypt analysis skills to evaluate the crypto-graphical technologies
3. Evaluate security mechanisms using rigorous approaches, including theoretical derivation, modeling, and simulations.
4. Formulate research problems in the computer security field.
5. Develop solutions to the formulated problems.

SECTION-I

Unit - 1: Overview of Cryptography **(8)**

Introduction, Information security and cryptography, Basic terminology and concepts, Symmetric-key encryption, Block Ciphers, DES ,AES Digital signatures, Public-key cryptography, Hash functions, Protocols and mechanisms, Key establishment, management, and certification, Pseudorandom numbers and sequences, Classes of attacks andsecurity models.

Unit – 2: Modular Arithmetic **(7)**

A quick introduction to groups, rings, integral domain and fields(including:Lagrange theorem, Structure of cyclic and abelian groups, isomorphism theorems), Probability theory, Information theory, Complexity theory, Number theory, Abstract algebra, Finite fields, The integer factorization problem,

Unit- 3: Cryptography and number theory **(7)**

The RSA problem, The Diffie-Hellman problem, Composite moduli.Elliptic Curve Cryptography

SECTION-II

Unit - 4: Mathematical Background **(8)**

Cyclotomic extensions, Fundamental theorem of Galois Theory, Geometric constructions and Galoistheory of Equations (Statement only of Abel Ruffini), Solving Cubic and Bi-quadratic polynomials using radicals. 8-Lectures

Unit - 5: Key Establishment Protocols **(8)**

Introduction, Key transport based on symmetric encryption, Key agreement based on symmetric techniques, Key transport based on public-key encryption, Key agreement based on asymmetric techniques, Secret sharing, Key Management Techniques, Techniques for distributing public keys, Techniques for controlling key usage, Key management involving multiple domains.

Unit - 6:WebServer Security: **(7)**

Physical security for servers, Host security for servers, securing web applications. Deploying SSL server certificates, securing your web service, computer crime Security for content providers: Controlling access to web content, Client-side digital certificates, code signing and Microsoft's Authenticode.

Internal Continuous Assessment (ICA):

Assignments : Minimum 5 to 6 assignments based on above topics.

Reference Book:

1. Alfred J. Menezes, Paul C. van Oorschot and Scott A.Vanstone,“Handbook of Applied Cryptography” CRC Press
2. Cryptography and Network Security: Principles and Practice(ISBN0131873164),4/e,by William Stallings
3. B.Schneier,“Applied Cryptography:Protocols,Algorithms and Source Code in C”,2nd Edition, John Wiley & Sons, 1995.
4. MattBishop,Computer Security:Art and Science, Addison-Wesley,2002.
5. Mihir Bellare and Phillip Rogaway,“Introduction to Modern Cryptography”
6. Fieldand GaloisTheoryBy Patrick Mor and iGTM
7. Fieldand GaloisTheory By J.M.Howie (Paperback)
8. GaloisTheory,Lecture notes by Emil Artin
9. GaloisTheory, TIFR Lecture notes
10. GaloisTheory by H.M. Edward(GTM)
11. Algebra By M.Artin

WALCHANDINSITUTEOFTECHNOLOGY

M.Tech.(Computer Science & Engineering) Part I, Semester I

CSP1CE13– CORE ELECTIVE-I: NATURAL LANGUAGE PROCESSING

Teaching Scheme	**Examination Scheme**
Lectures :3Hours/Week,3 Credits	**ESE–60Marks**
Practical : 2 Hours/Week, 1 Credit	**ISE –40 Marks**
	ICA–25 Marks

COURSE OUTCOMES:

At the end of the course students will be able to

1. Demonstrate the fundamental mathematical models and algorithms in the field of NLP.
2. Apply these mathematical models and algorithms in applications of software design and implementation for NLP.
3. Use tools to analyze language resource annotation and apply to data for acquiring intended information.
4. Design and implement various NLP applications.

SECTION-I

Unit – 1 : Introduction **(6)**

Introduction to NLP, Machine Learning and NLP, Biology of Speech Processing; Place and Manner of Articulation, Word Boundary Detection, Arg-Max Computation, Lexical Knowledge Networks.

Unit – 2 : Word-netTheory **(6)**

Semantic Roles , Word Sense Disambiguation (WSD) : Word-Net, Word-net Application in Query Expansion, Wiktionary, semantic relatedness , Measures of Word-Net Similarity, Similarity Measures. Resnick's work on Word-Net Similarity, Indian Language Word-nets and Multilingual Dictionaries, Multi-linguality, Metaphors, Co references

Unit – 3 : Theories of Parsing **(6)**

Parsing Algorithms, Evidence for Deeper Structure, Top Down Parsing Algorithms, Noun Structure, Non-noun Structure and Parsing Algorithms, Robust and Scalable Parsing on Noisy Text as in Web documents Probabilistic parsing, Hybrid of Rule Based and Probabilistic Parsing sequence labeling, Training issues, Arguments and Adjuncts, inside- outside probabilities, Scope Ambiguity and Attachment Ambiguity resolution.

Unit – 4 : Speech **(6)**

Phonetics , HMM, Morphology, Morphology fundamentals; Morphological Diversity of Indian Languages; Morphology Paradigms; Finite State Machine Based Morphology; Automatic Morphology Learning; Shallow Parsing; Named Entities; Maximum Entropy Models; Random Fields.

SECTION-II

Unit – 5 : Graphical Models **(6)**

Graphical Models for Sequence, Labelling in NLP, Consonants (place and manner of articulation) and Vowels , Forward Backward probability, Viterbi Algorithm

Unit - 6 : Phonology **(6)**

Sentiment Analysis and Opinions on the Web, Machine TranslationandMTTools- GIZA++ and Moses. Text Entailment , POS Tagging. ASR, Speech Synthesis , Precision, Recall, F-score, Map.

Unit - 7 : SemanticRelations **(6)**

UNL, Towards Dependency Parsing, Universal Networking Language, Semantic Role Extraction, Baum Welch Algorithm, HMM and Speech Recognition. HMM training, Baum Welch Algorithm; HMM training

Unit – 8 :Applications **(6)**
Sentiment Analysis;Text Entailment; Robust and Scalable Machine Translation;Question Answering in Multilingual Setting; Cross Lingual Information Retrieval (CLIR).

Internal Continuous Assessment (ICA):
It consists of minimum eight tutorials based upon each chapter of above curriculum. Tutorial shall include writing algorithms and implementing them.

Text Books:

1. Allen, James,“Natural Language Understanding”,Second Edition, Benjamin/Cumming,1995.
2. Charniack, Eugene,“Statistical Language Learning”, MITPress,1993.
3. Jurafsky, Danand Martin, James, Speech and Language Processing, Second Edition, Prentice Hall, 2008.
4. Manning, Christopher and Heinrich, Schutze, Foundations of Statistical Natural Language Processing, MIT Press, 1999.

Reference Books:

1. Jurafsky,D.,andMartin,J.H.(2008).“SpeechandLanguageProcessing”(2ndEdition). Upper Saddle River, NJ: Prentice Hall
2. Bird, S., Klein, E., Loper, E. (2009).“Natural Language Processing with Python”. Sebastopol, CA: O'Reilly Media.
3. Radford, Andrewet.al.,“Linguistics, An Introduction”,CambridgeUniversityPress,1999.

WALCHANDINSTITUTEOFTECHNOLOGY

M. Tech. (Computer Science & Engineering) Part I, Semester I

CSP1SM1 – SEMINAR

Teaching Scheme | **Examination Scheme**

Practicals:4 Hours/Week, 2 Credits | **ICA– 50 Marks**

Student must deliver a seminar based on the relevant topic based on recent research. The seminar topic shall be finalized in consultation with allotted supervisor. The seminar delivery shall be in front of the panel of PG faculty. The student must submit a seminar report on topic selected.

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR
F.Y.M.Tech.(Computer Science and Engineering), Semester-I
CMP1AC - YOGA FOR STRESS MANAGEMENT

Teaching Scheme:
Lectures- 2 Hrs/week

Examination Scheme :
ESE – 50 Marks

COURSE OBJECTIVES:-

1. To train the students in the fields of Yoga and Stress management
2. To promote awareness for positive health and personality development in the student through Yoga
3. To achieve overall Good Health of Body and Mind.
4. To increase the levels of happiness.

COURSE OUTCOMES:-

At the end of the course, students will be able to

1. Manage stress through breathing, awareness, meditation, and healthy movement.
2. Achieve overall health of body and mind by overcoming stress
3. Use their bodies in a healthy way and perform well in sports and academics.
4. Build concentration, confidence, and positive self-image.

CourseContents

Unit- 1: Stress & Yoga Philosophy **(3)**

Unit- 2: Managing Stress **(3)**

Unit- 3: Worthy Tips For Gracious Living **(3)**

Unit- 4: Panch Kosha, Trikaya & Abha(Five Sheaths, Three Bodies& Aura) **(3)**

Unit- 5: Antahkarana Chatusthya Aur Nigrah(Four-Fold Mind & its Control) **(3)**

Unit- 6: Introduction to Yoga **(3)**

Unit- 7: Conditioning The Mind **(3)**

Unit- 8: Pranayama (Breath Control) **(3)**

Unit- 9: Dhaarna, Dhyana & Samadhi (Concentration, Meditation & Super-Consciousness) **(3)**

TEXTBOOK:

1. AcharyaYatendra,Yoga&StressManagement,DivyaYogNiketanTrust,Finger-Print!Life,An imprint of Prakash Books India Pvt. Ltd., New Delhi

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REFERENCE BOOKS:

1. H R Nagendra and R Nagarathna. Yoga for Promotion of Positive Health. Swami Vivekananda Yoga Prakashana.
2. Contrada,R.,& Baum,A.(Eds.).The handbook of stress science: Biology, psychology and health. Springer Publishing Company.
3. Vanden Bergh,O. Principles and Practice of Stress Management. Guilford Publications.
4. Swami Muktibodhananda, Hatha Yoga Pradipika,, Bihar School ofYoga
5. Swami Satyananda Saraswati,Four Chapters on Freedom, Bihar School of Yoga
6. Swami Tapasyananda,Srimad Bhagavat Gita, Sri RamakrishnaMath

WALCHANDINSTITUTEOFTECHNOLOGY
M. Tech. (Computer Science & Engineering)Part I, Semester II
CSP2CC1 –ADVANCED CLOUD COMPUTING

Teaching Scheme	**Examination Scheme**
Lectures– 3Hours/week,3Credits	**ESE–60 Marks**
Practical – 2 Hour/week, 1 Credit	**ISE –40 Marks**
	ICA–25 Marks

COURSE OUTCOMES:

1. Compare the strengths and limitations of cloud computing
2. Identify the architecture, infrastructure and delivery models of cloud computing
3. Apply suitable virtualization concept.

SECTION I

Unit-I **(7)**

Overview of Computing Paradigm: Recent trends in Computing: Grid Computing, Cluster Computing, Distributed Computing, Utility Computing, Cloud Computing. Evolution of cloud computing: Business driver for adopting cloud computing.

Introduction to Cloud Computing :Cloud Computing (NIST Model): Introduction to Cloud Computing, History of Cloud Computing, Cloud service providers; Properties, Characteristics & Disadvantages: Pros and Cons of Cloud Computing, Benefits of Cloud Computing, Cloud computing vs. Cluster computing vs. Grid computing; Role of Open Standards

Unit-II **(6)**

Cloud Computing Architecture: Cloud computing stack: Comparison with traditional computing architecture (client/server), Services provided at various levels, How Cloud Computing Works, Role of Web services; Service Models (XaaS): Infrastructure as a Service(IaaS), Platform as a Service(PaaS), Software as a Service(SaaS);Deployment Models: Public cloud, Private cloud, Hybrid cloud, Community cloud.

Unit-III **(7)**

Infrastructure as a Service(IaaS): Introduction to IaaS, IaaS definition, Introduction to virtualization, Different approaches to virtualization, Hypervisors, Machine Image, Virtual Machine(VM).

Resource Virtualization: Server, Storage, Network, Virtual Machine(resource) provisioning and manageability, storage as a service, Data storage in cloud computing(storage as a service); Examples: Amazon EC2, Renting, EC2 Compute Unit, Platform and Storage, pricing, customers, Eucalyptus.

SECTIONII

Unit-IV **(7)**

Platform as a Service(PaaS):Introduction to PaaS: What is PaaS, Service Oriented Architecture (SOA), Cloud Platform and Management, Computation, Storage, Examples, Google App Engine, Microsoft Azure, SalesForce.com‟s Force.com platform.

Unit-V **(7)**

Software as a Service(PaaS):Introduction to SaaS, Web services, Web 2.0, Web OS, Case Study on SaaS. Service Management in Cloud Computing: Service Level Agreements (SLAs), Billing & Accounting, Comparing Scaling Hardware: Traditional vs. Cloud, Economics of scaling: Benefitting enormously, Managing Data, Looking at Data, Scalability & Cloud Services, Database & DataStores in Cloud, Large Scale Data Processing.

Unit-VI **(7)**

Cloud Security: Network level security, Host level security, Application level security, Data security and Storage: Data privacy and security Issues, Jurisdictional issues raised by Data location, Identity& Access Management, Access Control, Trust, Reputation, Risk, Authentication in cloud computing, Client access in cloud, Cloud contracting Model, Commercial and business

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Internal Continuous Assessment (ICA):

ICA shall be based upon 6 to 8 assignments based upon above syllabus.

Reference Books

1. Cloud Computing Bible,BarrieSosinsky,Wiley-India,2010
2. Cloud Computing Principals and Paradigms: Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, Wile, 2011
3. Cloud Computing: Principles, Applications, Editors: Nikos Systems and Antonopoulos, Lee Gillam, Springer, 2012
4. Cloud Security: A Comprehensive Guide to Secure Cloud Computing,RonaldL.Krutz,Russell Dean Vines, Wiley-India, 2010

WALCHAND INSTITUTE OF TECHNOLOGY
M. Tech. (Computer Science & Engineering) Part I, Semester II
CSP2CC2 - INTERNET ROUTING ALGORITHM

Teaching Scheme	**Examination Scheme**
Lectures–3 Hours/week, 3 Credits	**ESE–60 Marks**
	ISE–40 Marks

COURSE OUTCOMES:

On successful completion of the course, Student will be able to:

1. To comprehend basic concepts of network routing.
2. Demonstrate the working of various routing algorithms & Protocols
3. Describe the Router Architecture, its functions and types
4. Analyze and compare the network algorithms

SECTION I

Unit-1: Network Basics (7)

OSI Model, Network Hardware, Transmission media, Bridge, Router , Gateways, Network Software Components, MAC, Data Link Protocols, Switching Techniques TCP/IP Protocol suite.

Unit-2:Networking and Network Routing (7)

Addressing and Internet Service: An Overview, Network Routing, IP Addressing, Service Architecture, Protocol Stack Architecture, Router Architecture, Network, Topology, Architecture, Network Management Architecture, Public Switched Telephone Network

Unit-3 :Routing Algorithms (6)

Shortest Path and Widest Path: Bellman–Ford Algorithm and the Distance Vector Approach, Dijkstra's Algorithm, Widest Path Algorithm, Dijkstra-Based Approach, Bellman– FordBased Approach, k-Shortest Paths Algorithm. OSPF and IntegratedIS-IS:OSPF: Protocol Features,OSPF Packet Format, Integrated ISIS, Key Features, comparison BGP : Features, Operations, Configuration Initialization, phases, Message Format. IP Routing and Distance Vector Protocol Family : RIPv1 and RIPv2

SECTION-II

Unit-4: Routing Protocols (6)

Framework and Principles Routing Protocol, Routing Algorithm, and Routing Table, Routing Information Representation and Protocol Messages, Distance Vector Routing Protocol, Link State Routing Protocol, Path Vector Routing, Protocol, Link Cost.

Unit-5:Internet Routing and Router Architectures (6)

Architectural View of the Internet, Allocation of IP Prefixes and AS Number, Policy-Based Routing, Point of Presence, Traffic Engineering Implications, Internet Routing Instability.

Router Architectures: Functions, Types, Elements of a Router, Packet Flow, Packet Processing: Fast Path versus Slow Path, Router Architectures

Unit-6: Analysis of Network Algorithms (6)

Network Bottleneck, Network Algorithmics, Strawman solutions, Thinking Algorithmically, Refining the Algorithm, Cleaning up, Characteristics of Network Algorithms. IP AddressLookup Algorithms : Impact, Address Aggregation, Longest Prefix Matching, Naïve Algorithms, Binary, Multibit and Compressing Multibit Tries, Search by Length Algorithms, Search by Value Approaches, Hardware Algorithms, Comparing Different Approaches IP Packet Filtering and Classification : Classification, Classification Algorithms, Naïve Solutions, Two-Dimensional Solutions, Approaches for d Dimensions,

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Text Books:

1. Network Algorithmics: An Interdisciplinary Approach to Designing Fast Networked Devices George Varghese (Morgan Kaufmann Series in Networking)

Reference Book:

1. Network Routing: Algorithms, Protocols and Architectures Deepankar Medhi and Karthikeyan Ramasamy (Morgan Kaufmann Series in Networking)

WALCHAND INSTITUTE OF TECHNOLOGY, SOLAPUR

M. Tech. (Computer Science & Engineering) Part-I, Semester-II

CSP2CC3 : ADVANCED DATABASE SYSTEMS

Teaching Scheme	**Examination Scheme**
Lectures: 3 Hours/week, 3 Credits	**ESE:60Marks**
Practical : 2 Hours/week, 1 Credit	**ISE : 40 Marks**
	ICA :25 Marks

COURSE OUTCOMES:

1. Select the appropriate high-performance database like parallel and distributed database.
2. Infer and represent the real-world data using object-oriented database.
3. Interpret rule set in the database to implement data warehousing of mining.
4. Discover and design database for recent applications database for better interoperability.

SECTION I

Unit-1 **(8)**

Physical database design & Tuning Database workloads, physical design and tuning decisions, Need for Tuning Index selection: Guideline for index selection, Clustering & Indexing Tools for index selection Database Tuning: Tuning indexes, Tuning Conceptual schema Tuning Queries & views, Impact of Concurrency, Benchmarking

Unit-2 **(6)**

Distributed Databases Introduction, Design Framework, Design of database fragmentation, The Allocation of Fragments, Translation of global queries to fragment queries, Optimization of access queries, Distributed Transaction Management, Concurrency Control, Reliability.

Unit-3 **(8)**

Advance Transaction Processing Transaction Processing Monitors, Transactional Workflow, Real time transaction System, Long duration Transactions, Transaction Management in Multi-databases, Distributed Transaction Management, Main Memory Databases, and Advanced Transaction Models.

SECTION-II

Unit-4 **(8)**

Semi-Structured Data and XML Semi-Structured Data, Introduction to XML, Components of XML, XML schemas& DTD, Parsing XML, Xpath, XSLT, XQuery, Storage of XML data XML Technologies& Application: DOM &SAX Interfaces, XHTML, SOAP, WSDL, UDDI, XML database Application.

Unit-5 **(6)**

Emerging Trends in Databases: Temporal databases, Spatial & geographic databases, Multimedia Databases, Mobile Databases

Unit-6 **(8)**

Large-scale Data Management with HADOOP, Semi structured database COUCHDB: Introduction, Architecture and principles, features

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Internal Continuous Assessment (ICA):

Minimum 5-6 assignments on above mentioned syllabus.

Text Books:

1. Database system Concept by Silberschatz and Korth 6th Edition
2. Database Management System-Ramakrishna Gherkin(McGraw Hill)
3. Distributed Database Principals and systems-Stephanceri, Giuseppe Pelagatti.(McGraw Hill)

Reference Books:

1. Web Data Management, Abiteboul, Loana, Philippe Et.al Cambridge publication
2. Database Systems, Thomas Connolly, Carolyn Begg, Pearson 4th Edition
3. Database Management Systems by Raghu Ramakrishnan and Johannes Gehrke

WALCHAND INSITUTE OF TECHNOLOGY

M.Tech. (Computer Science & Engineering) Part I, Semester II

CSP2CE21 –CORE ELECTIVE II: MACHINE LEARNING

Teaching Scheme	**Examination Scheme**
Lectures :3 Hours/Week, 3 Credits	**ESE–60 Marks**
Practicals : 2 Hours/Week, 1 Credit	**ISE –40 Marks**
	ICA–25 Marks

COURSEOUTCOMES:

At the end of the course students will be able to

1. Demonstrate types of machine learning algorithms.
2. Design a model by selecting appropriate machine learning algorithm for a given problem.
3. Validate designed machine learning model.
4. Evaluate and tune machine learning model based on various parameters.
5. Design various applications using machine learning algorithm.

SECTION-I

Unit-1: Introduction **(6)**

Machine learning: what and why? Supervised learning, Unsupervised learning, Some basic concepts in machine learning, Definition of learning systems. Goals and applications of machine learning. Aspects of developing a learning system: training data, concept representation, function approximation.

(Chapter1 from Book1,Chapter1 from Book 2)

Unit-2: Linear and Logistic Regression **(8)**

Linear regression: Introduction, Model specification, Maximum likelihood estimation (least squares), Robust linear regression, Ridge regression, Bayesian linear regression Logistic regression: Introduction, Model specification, Model fitting, Bayesian logistic regression, Online learning and stochastic optimization, Generative vs. discriminative classifiers.

(Chapter7and 8 from Book 2)

Unit -3: Decision Tree Learning and Ensemble Methods **(8)**

Representing concepts as decision trees. Recursive induction of decision trees. Picking the best splitting attribute: entropy and information gain. Searching for simple trees and computational complexity. Occam'sraz or.Overfitting, noisy data and pruning, Ensemble Methods: Bagging and Boosting

(Chapter 3 from Book 1,Chapter 14 from Book 3)

SECTION-II

Unit-4: Clustering **(6)**

Introduction, Dirichlet process mixture models, Affinity propagation, Spectral clustering, Hierarchical clustering, Clustering data points and features.*(Chapter 25 from Book 2)*

Unit-5: Sparse Kernel Machines **(8)**

Introduction to Support Vector Machines(SVM), Maximum Margin Classifiers, Relevance Vector Machines. Applications of Support Vector Machines. *(Chapter 7 from Book 3)*

Unit-6: Neural Networks and Deep Learning **(8)**

Feed-forward Network Functions, Network Training, Error Backpropagation, Regularization in Neural Networks, Deep Learning: Introduction, Deep Neural Networks, Applications of Deep Networks.

(Chapter 5 from Book 3, Chapter28 from Book2)

Unit-7: Key Ideas in Machine Learning **(4)**

Introduction, Key Perspectives on Machine Learning, Key Results.Where is Machine Learning Headed Next?
(Chapter 14 of upcoming 2^{nd} Edition of Book 1)

Internal Continuous Assessment(ICA):

ICA shall be based upon minimum 6 laboratory experiment based upon above curriculum.

Reference Books:

1. Book 1: Machine Learning by Tom Mitchell, McGraw Hill (1stEdition)+New chapters from the upcoming second edition.
2. Draft content of chapter 14 of upcoming 2^{nd} Edition of Book 1
   http://www.cs.cmu.edu/~tom/mlbook/keyIdeas.pdf
3. Book 2: Machine Learning: A Probabilistic Perspective by Kevin Patrick Murphy
4. Book 3: Pattern Recognition and Machine Learning (Information Science and Statistics) by Christopher M. Bishop

WALCHAND INSITUTE OF TECHNOLOGY

M.Tech. (Computer Science & Engineering) Part I, Semester II

CSP2CE22–CORE ELECTIVE-II: BLOCKCHAIN TECHNOLOGY

Teaching Scheme	**Examination Scheme**
Lectures :3 Hours/Week, 3 Credits	**ESE–60Marks**
Practical: 2 Hour/Week, 1 Credit	**ISE –40 Marks**
	ICA–25 Marks

COURSE OUTCOMES:

At the end of this course students will be able to–

1. Describe the basic concepts and technology used for blockchain.
2. Describe the primitives of the distributed computing and cryptography related to blockchain.
3. Illustrate the concepts of Bitcoin and their usage.
4. Implement Ethereum blockchain contract.
5. Apply security features in blockchain technologies.
6. Use smart contract in real world applications.

SECTION-I

Unit I: Basics **(8)**

Distributed Database, Two General Problem, Byzantine General problem and Fault Tolerance, Hadoop Distributed File System, Distributed Hash Table, ASIC resistance, Turing Complete. Cryptography: Hash function, Digital Signature - ECDSA, Memory Hard Algorithm, Zero Knowledge Proof.

Unit II: Blockchain **(7)**

Introduction, Advantage over conventional distributed database, Blockchain Network, Mining Mechanism, Distributed Consensus, Merkle Patricia Tree, Gas Limit, Transactions and Fee, Anonymity, Reward, Chain Policy, Life of Blockchain application, Soft & Hard Fork, Private and Public blockchain.

Unit III: Distributed Consensus **(7)**

Nakamoto consensus, Proof of Work, Proof of Stake, Proof of Burn, Difficulty Level, Sybil Attack, Energy utilization and alternate.

SECTION-II

Unit IV: Cryptocurrency: **(8)**

History, Distributed Ledger, Bitcoin protocols - Mining strategy and rewards, Ethereum - Construction, DAO, Smart Contract, GHOST, Vulnerability, Attacks, Sidechain, Namecoin

Unit V: CryptocurrencyRegulation: **(8)**

Stakeholders, Roots of Bit coin, Legal Aspects-Crypto currency Exchange, Black Market and Global Economy.

Unit VI: Applications **(7)**

Internet of Things, Medical Record Management System, Domain Name Service and future of Blockchain.

Internal Continuous Assessment (ICA):

Assignments: Minimum 5 to 6 assignments based on above topics.

Text Book :

1. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction, Princeton University Press (July 19, 2016).

Reference Books:

1. Antonopoulos, Mastering Bitcoin: Unlocking Digital Cryptocurrencies
2. Satoshi Nakamoto,Bitcoin: A Peer-to-Peer Electronic Cash System
3. Dr. Gavin Wood, "ETHEREUM: A Secure Decentralized Transaction Ledger,"Yellowpaper.2014.
4. Nicola Atzei, Massimo Bartoletti and Tiziana Cimoli, A survey of attacks on Ethereum smart contracts

WALCHAND INSITUTE OF TECHNOLOGY

M.Tech. (Computer Science & Engineering) Part I, Semester II

CSP2CE23 – CORE ELECTIVEII: REINFORCEMENT LEARNING

Teaching Scheme	**Examination Scheme**
Lectures :3 Hours/Week, 3 Credits	**ESE–60 Marks**
Practical: 2 Hours/Week, 1 Credit	**ISE –40 Marks**
	ICA–25 Marks

COURSE OUTCOMES:

At the end of the course students will be able to

1. Demonstrate basic and advanced reinforcement learning techniques.
2. Identification of suitable learning tasks to which reinforcement learning techniques can be applied.
3. Describe some of the current limitations of reinforcement learning techniques.
4. Formulate decision problems based on various parameters to evaluate models.

SECTION I

Unit-1: Introduction **(5)**

Reinforcement Learning, Examples, Elements of Reinforcement Learning An Extended Example: Tic-Tac-Toe, History of Reinforcement Learning

Unit-2: Evaluative Feedback **(5)**

Ak-armed Bandit Problem, Action-value Methods, The 10-armed Testbed, Incremental Implementation

Unit-3: The Reinforcement Learning Problem **(6)**

The Agent–Environment Interface, Goals and Rewards, Returns, Unified Notation for Episodic and Continuing Tasks, Value Functions, Optimal Value Functions, Optimality and Approximation

Unit-4: Finite Markov Decision Processes **(6)**

The Agent–Environment Interface, Goals and Rewards, Returns and Episodes, Unified Notation for Episodic and Continuing Tasks, Policies and Value Functions.

SECTIONII

Unit-5: Dynamic Programming **(5)**

Policy Evaluation (Prediction), Policy Improvement, Policy Iteration, Value Iteration, Asynchronous Dynamic Programming, Generalized Policy Iteration, Efficiency of Dynamic Programming, Introduction to Monte Carlo Methods

Unit-6: Temporal-Difference Learning **(5)**

TD Prediction, Advantages of TD Prediction Methods, Optimality of TD(0), Sarsa: On-policy TD Control, Q-learning: Off-policy TD Control, Games, After states and Other Special Cases

Unit-7: Planning and Learning **(6)**

Models and Planning, Dyna: Integrating Planning, Acting and Learning, When the Model Is Wrong, Prioritized Sweeping, Expected vs. Sample Updates, Trajectory Sampling, Real-time Dynamic Programming, Planning at Decision Time, Heuristic Search, Rollout Algorithms.

Unit-8: Applications and Case Studies **(6)**

TD-Gammon, Samuel"s Checkers Player, Watson"s Daily-Double Wagering, Optimizing Memory Control, Human-level Video Game Play, Mastering the Game of Go, AlphaGo, AlphaGo Zero, Personalized Web Services, Thermal Soaring.

Internal Continuous Assessment(ICA):
Minimum 5-6 assignments on above syllabus.

Text Books:

1. Reinforcement Learning: An Introduction (Second edition+ Upcoming Edition) by : Richard S. Sutton and Andrew G.Barto, MIT Press Publication

(The book is available at http://incompleteideas.net/book/the-book-2nd.html
Upcoming edition's January 1 2018 draft available at
http://incompleteideas.net/book/bookdraft2018jan1.pdf

Reference Books:

1. Reinforcement Learning: With OpenAI, Tensor Flow and Keras Using Python By Abhishek Nandy, Manisha Biswas. Apress Publication
2. Reinforcement Learning:State-of-the-Art,Marco Wiering and Martijnvan Otterlo, Eds.
3. Artificial Intelligence:A Modern Approach, Stuart J. Russell and Peter Norvig.
4. Deep Learning, Ian Goodfellow,Yoshua Bengio and Aaron Courville.

WALCHAND INSTITUTE OF TECHNOLOGY
M. Tech. (Computer Science & Engineering) Part I, Semester II
CSP2CE31– CORE ELECTIVE – III : DATA SCIENCE

Teaching Scheme	**Examination Scheme**
Lectures: 3 Hours/Week, 3 Credits	**ESE–60 Marks**
	ISE–40Marks

COURSE OUTCOMES:

At the end of the course the student will be able to:

1. Define data science and its fundamentals.
2. Demonstrate the process in data science.
3. Explain machine learning algorithms necessary for data sciences
4. Illustrate the process of feature selection and analysis of data analysis algorithms.
5. Visualize the data and follow of ethics.

SECTION-I

Unit-1: Introduction (7)

Introduction: What is Data Science? Big Data and Data Science hype– and getting past the hype, Why now? – Data fication, Current landscape of perspectives, Skillsets. Needed Statistical Inference: Populations and samples, Statistical modeling, probability distributions, fitting a model, -Introduction to R

Unit-2: Exploratory Data Analysis (7)

Exploratory Data Analysis and the Data Science Process: Basic tools (plots, graphs and summary statistics) of EDA, Philosophy of EDA. The Data Science Process, Case Study: Real Direct (on line real estate firm). Three Basic Machine Learning Algorithms: Linear Regression, k-Nearest Neighbors (k-NN), k-means

Unit-3: Machine Learning Algorithms (7)

One More Machine Learning Algorithm and Usage in Applications: Motivating application: Filtering Spam, Why Linear Regression and k-N Nare poor choices for Filtering Spam, Naïve Bayes and why it works for Filtering Spam,Data Wrangling: APIs and other tools for scrapping the Web

SECTION-II

Unit-4: Feature Generation (8)

Feature Generation and Feature Selection (Extracting Meaning from Data): Motivating application : user (customer) retention. Feature Generation (brainstorming, role of domain expertise, and place for imagination), Feature Selection algorithms. Filters; Wrappers; Decision Trees; Random Forests.

Unit-5: Recommendation Systems (6)

Recommendation Systems: Building a User-Facing Data Product, Algorithmic ingredients of a Recommendation Engine, Dimensionality Reduction, Singular Value Decomposition, Principal Component Analysis, Exercise: build your own recommendation system

Unit-6: Data Visualization (9)

Mining Social-Network Graphs: Social networks as graphs, Clustering of graphs, Direct discovery of communities in graphs, Partitioning of graphs, Neighbourhood properties in graphs, Data Visualization: Basic principles, ideas and tools for data visualization. Data Science and Ethical Issues, Discussions on privacy, security, ethics, Next-generation data scientists

Text books:

1. Doing Data Science, Cathy O‟Neil and Rachel Schutt, Straight Talk from The Frontline. O‟Reilly, 2014
2. Mining of Massive Datasets. V2.1 Jure Leskovek, Anand Rajaraman and Jeffrey Ullman, Cambridge University Press, 2014

Reference Books

1. Data Mining: Concepts and Techniques Jiawei Han, Micheline Kamber and JianPei, Third Edition, 2012.
2. Machine Learning: A Probabilistic Perspective, Kevin P.Murphy,2013

CSP2CE32–CORE ELECTIVE–III: WIRELESS SENSOR NETWORK

Teaching Scheme | **Examination Scheme**
Lectures– 3 Hours/week, 3 Credits | **ESE–60 Marks**
ISE–40 Marks

COURSE OUTCOMES:
1. Explain the basic concepts of wireless sensor networks, sensing, computing and communication tasks.
2. Describe standards and communication protocols adopted in wireless sensor networks.
3. Demonstrate knowledge of MAC and routing protocols developed for WSN.

SECTION-I

Unit-1: Introduction to WSN (8)
Introduction to WSN, Basic Overview of the Technology, Basic Sensor Network Architectural Elements, Brief Historical Survey of Sensor Networks, Challenges and Hurdles, Applications of Wireless Sensor Networks, Range of Applications, and Category 1 WSN Applications: Sensor and Robots, Reconfigurable Sensor Networks, Highway Monitoring, Wildfire Instrumentation, Nanoscopic Sensor Applications, Habitat Monitoring, Category 2 WSN Applications: Home Control, Building Automation, Industrial Automation, Medical Applications

Unit-2: Basic Wireless Sensor Technology (8)
Introduction, Sensor Node Technology, Hardware and Software, Sensor Taxonomy, WN Operating Environment, WN Trends, Introduction to Wireless Transmission Technology and Systems, Radio Technology Primer, Propagation and Propagation Impairments, Modulation, Available Wireless Technologies, Campus Applications, MAN/WAN Applications

Unit-3: Factors Influencing WSN Design (4)
Hardware Constraints, Fault Tolerance, Scalability, Production Costs, WSN Topology: Pre-deployment and Deployment Phase, Post-deployment Phase, Re-deployment Phase of Additional Nodes, Transmission Media, Power Consumption, Sensing, Data Processing, Communication

SECTION-II

Unit-4: Physical layer (6)
Introduction to Wireless channel and communication fundamentals :Frequency allocation, Modulation and demodulation, Wave propagation effects and noise, Channel models, Spread-spectrum communications, Packet transmission and synchronization, Quality of wireless channels and measures for improvement, Physical layer and transceiver design considerations in WSNs: Energy usage profile, Choice of modulation scheme, Dynamic modulation scaling, Antenna considerations

Unit-5: MAC protocols and Link-layer protocols (4)
Fundamentals of (wireless) MAC protocols, Low duty cycle protocols and wakeup concepts, Contention-based protocols, Schedule-based protocols, The IEEE802.15.4 MAC protocol, Link-layer protocols :Fundamentals: tasks and requirements, Error control, Framing, Link management

Unit-6: Network Layer and Transport Layer (8)
Challenges for Routing, Data-centric and Flat-Architecture Protocols, Hierarchical Protocols, Geographical Routing Protocols, QoS-Based Protocols Transport Layer, Challenges for Transport Layer, Reliable Multi-Segment Transport (RMST) Protocol, Pump Slowly, Fetch Quickly (PSFQ) Protocol, Congestion Detection and Avoidance(CODA) Protocol, Event-to-Sink Reliable

Transport (ESRT) Protocol, GARUDA, Real-Time and Reliable Transport Protocol

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Internal Continuous Assessment (ICA):

Minimum 5-6 assignments on above mentioned syllabus.

Text Books:

1. Protocols &Architectures for Wireless Sensor Networks by Holger Karl,AndreasWillig,Wiley,2005
2. Wireless Sensor Networks: Technology, Protocols and Applications by Kazem Sohraby, Daniel Minoli, Taieb Znati
3. Wireless Sensor Networks by Ian F. Akyildiz, Mehmet Can Vuran, A John Wiley and Sons,Ltd. Publication

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Reference Books:

1. Wireless sensor networks Edited by C.S.Raghavendra, Pub:Springer
2. Carlos de Morais Cordeiroand Dharma Prakash Agrawal,"AdHoc and Sensor Networks :Theory and Applications", Second Edition, World Scientific Publishers, 2011
3. Jagannathan Sarangapani, Wireless Adhoc and Sensor Networks: Protocols, Performance, and Control, CRC Press, 2007

WALCHANDINSTITUTEOFTECHNOLOGY

M. Tech. (Computer Science & Engineering) Part I, Semester II

CSP2CE33–COREELECTIVE–III:HIGHPERFORMANCECOMPUTING

Teaching Scheme	**Examination Scheme**
Lectures–3 Hours/week, 3 Credits	**ESE–60 Marks**
	ISE–40 Marks

COURSE OUTCOMES:

At the end of the course, students will be able to:

1. Demonstrate the role of HPC in science and engineering.
2. Design the concept of parallel computing paradigm.
3. Analyze the functionality of Multi-Processor.
4. Learn importance of interconnection networks used to achieve high performance in modern system.

SECTIONI

Unit-1 : Introduction to Parallel Processing (7)

Levels of Parallelism (instruction, transaction, task, thread, memory, and function), Models (SIMD, MIMD, SIMT, SPMD, Data Flow Models, Demand-driven Computation). HPC Platforms: Message-passing interface (MPI),Shared-memory thread-based OpenMP programs, hybrid(MPI/Open MP) programs, Grid Computing, Cloud Computing , Multi- Core Processors, accelerators, GPGPUs.

Unit-2: Parallel Programs (8)

The Parallelization Process: Steps in the process, Parallelizing computation versus data, Goals o fthe Parallelization Process, Parallelization of an Example Program: The equation solver kernel, Decomposition, Assignment, Orchestration: under the data parallel model, under the shared address space model and under the message passing model.

Unit-3: Parallel Models, Languages and Compilers (8)

Parallel Programming Models: Shared Variable Model, Message Passing Model, Data Parallel Model, Object Oriented Model and Functional and Logic Models. Parallel Languages and Compilers: Language Features for Parallelism, Parallel Language Constructs, Optimizing Compilers for Parallelism. Loop Parallelism and Pipelining: Loop transaction theory, Parallelization and wavefronting, Tiling and Localization and Software Pipelining.

SECTIONII

Unit-4: Parallel Program Development and Environment (7)

Parallel Programming Environments: Software tools and environments, Y-MP, Paragon and CM-5 Environment, Visualization and Performance Tuning. Synchronization and Multi-processing Models: Principles of Synchronization, Multiprocessor Execution Models, Multitasking on Cray Multiprocessors. Shared Variable Program Structures: Lock for protected access, Semaphores and Applications, Monitors and Applications.

Unit-5: Shared Memory Multiprocessor (8)

Cache Coherence: The Cache Coherence Problem, Cache Coherence through Bus Snooping. Memory Consistency: Sequential Consistency, Sufficient Conditions for Preserving Sequential Consistency. Synchronization: Components of a Synchronization event, Role of the user and Program, Mutual exclusion, Point to Point Event Synchronization, Global(Barrier) Event Synchronization.

Unit-6: Interconnection Network Design **(7)**

Basic Communication Performance: Latency, Bandwidth. Organizational Structure: Links, Switches and Network Interface. Interconnection Topology: Fully connected network, Linear array and rings, Multidimensional Meshes and Tori, Trees, Butterflies and Hypercube. Routing: Routing Mechanisms, Deterministic Routing, Turn-Model Routing and Adaptive Routing.

Text Books:

1. Ananth Grama, Anshul Gupta, George Karypis,Vipin Kumar,“Introduction to Parallel Computing”, Pearson Education, Second Edition, 2007.
2. Kai Hwang,Naresh Jotwani,“Advanced Computer Architecture: Parallelism, Scalability, Programmability”, McGraw Hill, Second Edition, 2010.
3. David Culler Jaswinder Pal Singh, "Parallel Computer Architecture: A hardware/Software Approach", Morgan Kaufmann, 1999.

Reference Books:

1. KaiHwang,,"ScalableParallelComputing",McGrawHill1998.
2. GeorgeS.AlmasiandAlanGottlieb,"HighlyParallelComputing",TheBenjaminandCummings Pub. Co., Inc
3. Georg Hager, Gerhard Wellein, “Introduction to High Performance Computing for Scientists and Engineers", Chapman & Hall / CRC Computational Science series, 2011.
4. Michael J. Quinn,“Parallel Programming in C with MPI and OpenMP”,McGraw- Hill International Editions, Computer Science Series, 2008.

WALCHAND INSTITUTE OF TECHNOLOGY

M.Tech. (Computer Science & Engineering) Part-I, Semester-II

CSP2SM2 – SEMINAR-II

Teaching Scheme:	**Examination Scheme**
Practical: 6 Hours/week, 3 Credits	**ICA– 100 Marks**

Student must deliver a seminar based on the relevant topic based on recent research. The seminar topic shall be finalized in consultation with allotted supervisor. The seminar delivery shall be in front of the panel of PG faculty.

The student must submit a seminar report on topic selected.